

Find Solution using North-West Corner method

	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	

	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	

TOTAL number of supply constraints : 3

TOTAL number of demand constraints : 4

	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	

$$\text{Total Supply} = \sum_{i=1}^3 a_i = 7 + 9 + 18 = 34$$

$$\text{Total Demand} = \sum_{j=1}^4 b_j = 5 + 8 + 7 + 14 = 34$$

$$\text{Total Supply} = \text{Total Demand} = 34$$

Problem is Balanced

1. North-West Corner method

First Allocation

	D_1	D_2	D_3	D_4	Supply
S_1	19 (5)	30	50	10	7 (2)
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5 (0)	8	7	14	

Move horizontally,

The rim values for $S_1=7$ and $D_1=5$ are compared.

1. North-West Corner method

First Allocation

	D_1	D_2	D_3	D_4	Supply
S_1	19 (5)	30	50	10	7 (2)
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5 (0)	8	7	14	

Move horizontally,

The rim values for $S_1=7$ and $D_1=5$ are compared.

The smaller of the two i.e. $\min(7,5) = 5$ is assigned to (S_1, D_1)

1. North-West Corner method

Second Allocation

	D_1	D_2	D_3	D_4	Supply
S_1	19 (5)	30 (2)	50	10	7 (2) (0)
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5 (0)	8 (6)	7	14	

Move horizontally,

The rim values for $S_1=2$ and $D_2=8$ are compared.

The smaller of the two i.e. $\min(2,8) = 2$ is assigned to (S_1, D_2)

This meets the complete demand of S_1 and leaves $8-2=6$ units with D_2

1. North-West Corner method

Third Allocation

	D_1	D_2	D_3	D_4	Supply
S_1	19 (5)	30 (2)	50	10	7 (2) (0)
S_2	70	30 (6)	40	60	9 (3)
S_3	40	8	70	20	18
Demand	5 (0)	8 (6) (0)	7	14	

Move vertically,

The rim values for $S_2=9$ and $D_2=6$ are compared.

The smaller of the two i.e. $\min(9,6) = 6$ is assigned to (S_2, D_2)

This meets the complete demand of D_2 and leaves $9-6=3$ units with S_2

1. North-West Corner method

Fourth Allocation

	D_1	D_2	D_3	D_4	Supply
S_1	19 (5)	30 (2)	50	10	7 (2) (0)
S_2	70	30 (6)	40 (3)	60	9 (3) 0
S_3	40	8	70	20	18
Demand	5 (0)	8 (6) (0)	7 (4)	14	

Move horizontally,

The rim values for $S_2=3$ and $D_3=7$ are compared.

The smaller of the two i.e. $\min(3,7) = 3$ is assigned to (S_2, D_3)

This meets the complete demand of S_2 and leaves $7-3=4$ units with D_3

1. North-West Corner method

Fifth Allocation

	D_1	D_2	D_3	D_4	Supply
S_1	19 (5)	30 (2)	50	10	7 (2) (0)
S_2	70	30 (6)	40 (3)	60	9 (3) (0)
S_3	40	8	70 (4)	20	18 (14)
Demand	5 (0)	8 (6) (0)	7 (4) 0	14	

Move vertically,

The rim values for $S_2=18$ and $D_3=4$ are compared.

The smaller of the two i.e. $\min(18,4) = 4$ is assigned to (S_3, D_3)

This meets the complete demand of D_3 and leaves $18-4=14$ units with S_3

1. North-West Corner method

Sixth Allocation

	D_1	D_2	D_3	D_4	Supply
S_1	19 (5)	30 (2)	50	10	7 (2) (0)
S_2	70	30 (6)	40 (3)	60	9 (3) 0
S_3	40	8	70 (4)	20	18 (14)
Demand	5 (0)	8 (6) (0)	7 (4) 0	14	

Move vertically,

The rim values for $S_2=14$ and $D_4=14$ are compared.

The smaller of the two i.e. $\min(14,14) = 14$ is assigned to $(S_3, D_4)=14$

1. North-West Corner method

Initial feasible solution is

	D_1	D_2	D_3	D_4	Supply
S_1	19 (5)	30 (2)	50	10	7 (2) (0)
S_2	70	30 (6)	40 (3)	60	9 (3) 0
S_3	40	8	70 (4)	20 (14)	18 (14) (0)
Demand	5 (0)	8 (6) (0)	7 (4)	14 (0)	

The total Transportation cost= **$19 \times 5 + 30 \times 2 + 30 \times 6 + 40 \times 3 + 70 \times 4 + 20 \times 14 = 1015$**

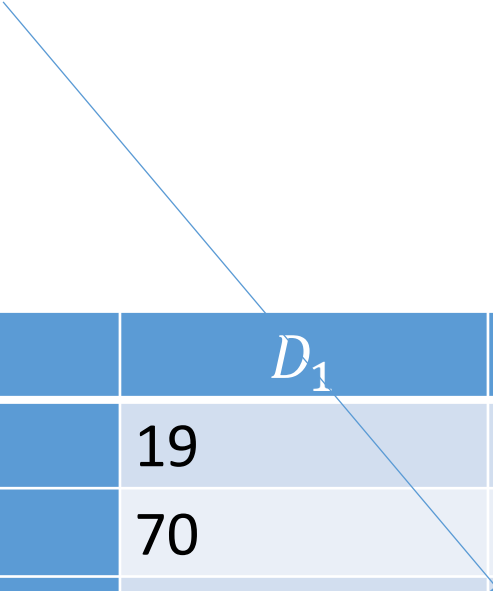
Least Cost method

	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	

Total Supply = Total Demand = 34

Problem is Balanced

Least Cost

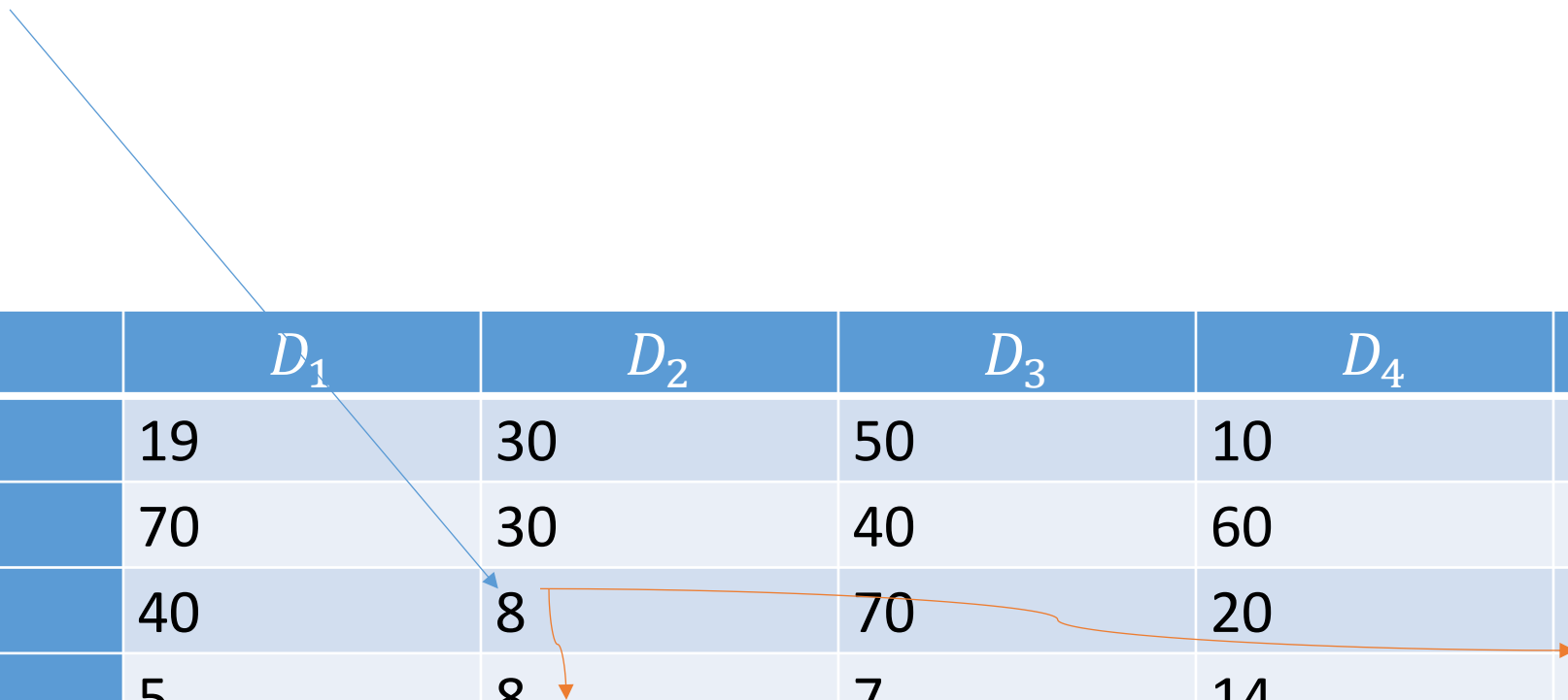


	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	

The least cost is 8 corresponds to the cells (S_3, D_2)

Allocate $\min(18, 8) = 8$ units to this cell.

Least Cost

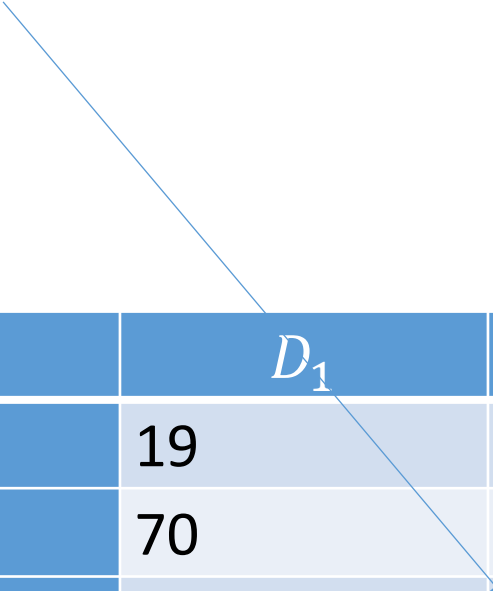


	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	

The least cost is 8 corresponds to the cells (S_3, D_2)

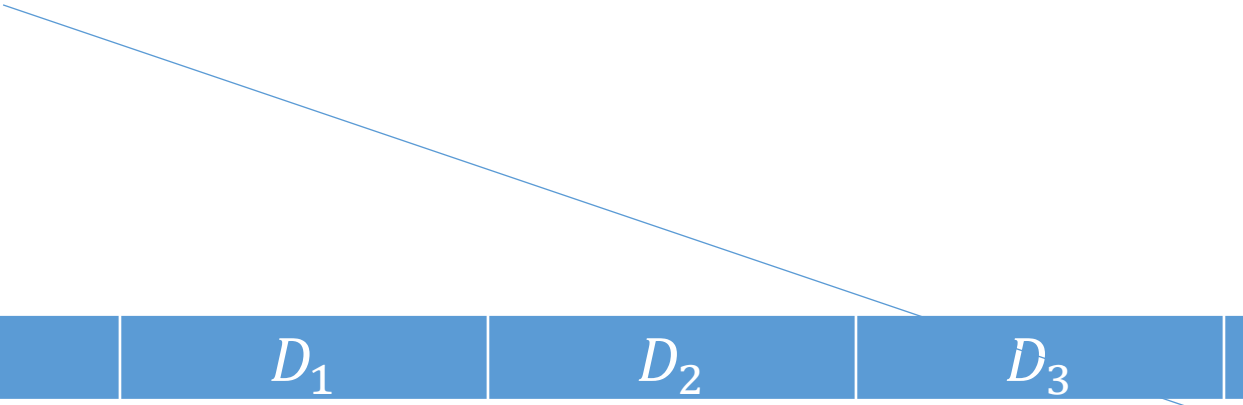
Allocate $\min(18, 8) = 8$ units to this cell.

Least Cost



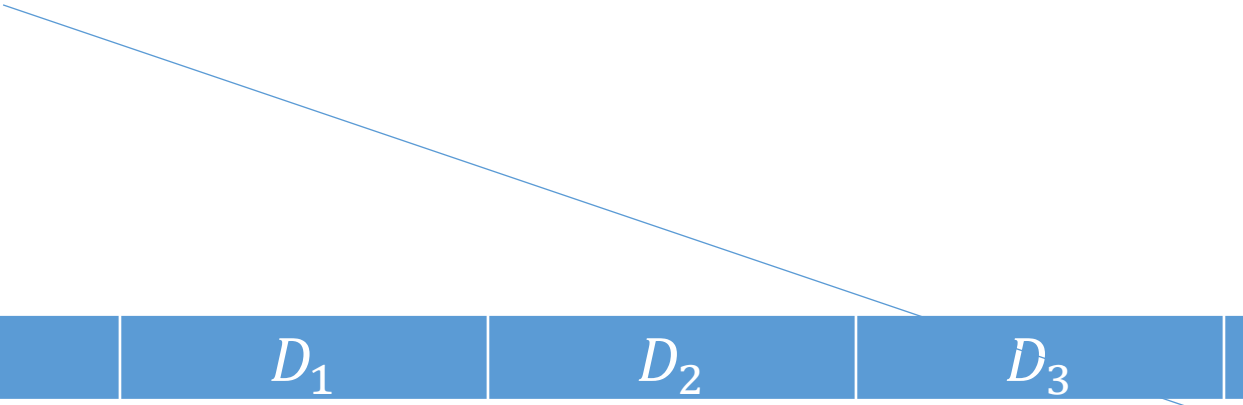
	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8 (8)	70	20	18 (10)
Demand	5	8 (0)	7	14	

Least Cost



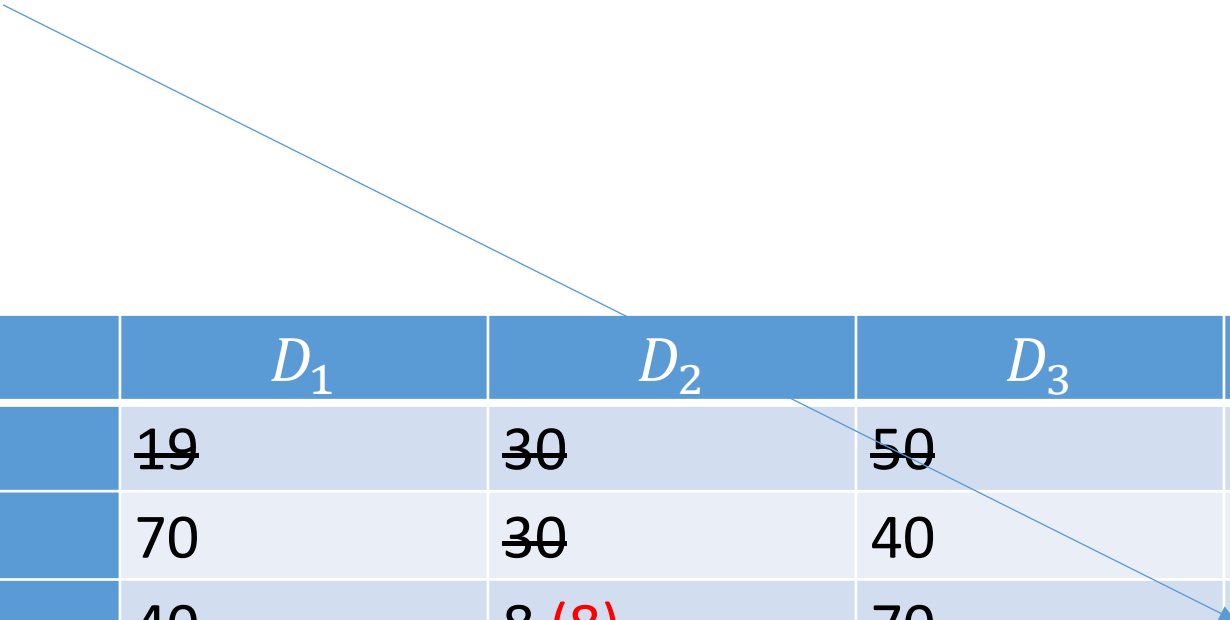
	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8 (8)	70	20	18 (10)
Demand	5	8 (0)	7	14	

Least Cost



	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10 (7)	7 (0)
S_2	70	30	40	60	9
S_3	40	8 (8)	70	20	18 (10)
Demand	5	8 (0)	7	14 (7)	

Least Cost

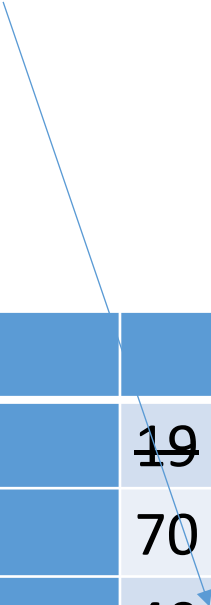


	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10 (7)	7 (0)
S_2	70	30	40	60	9
S_3	40	8 (8)	70	20	18 (10)
Demand	5	8 (0)	7	14 (7)	

Least Cost

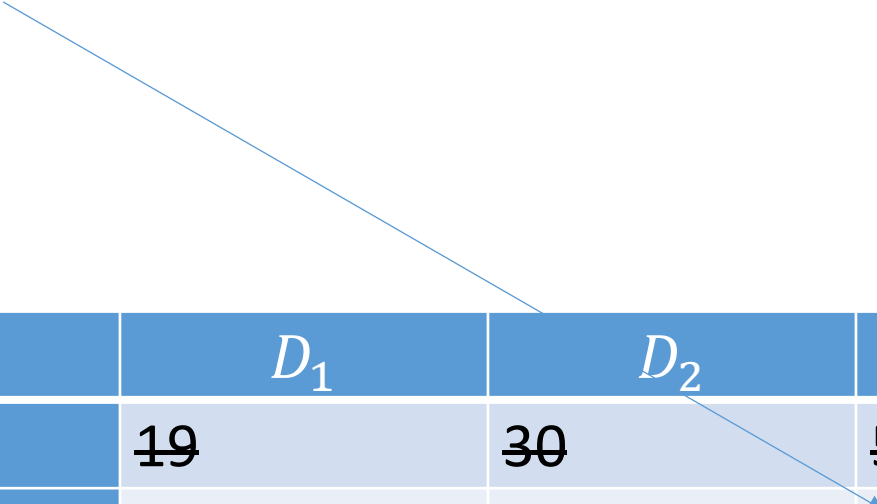
	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10 (7)	7 (0)
S_2	70	30	40	60	9
S_3	40	8 (8)	70	20 (7)	18 (10) (3)
Demand	5	8 (0)	7	14 (7) 0	

Least Cost



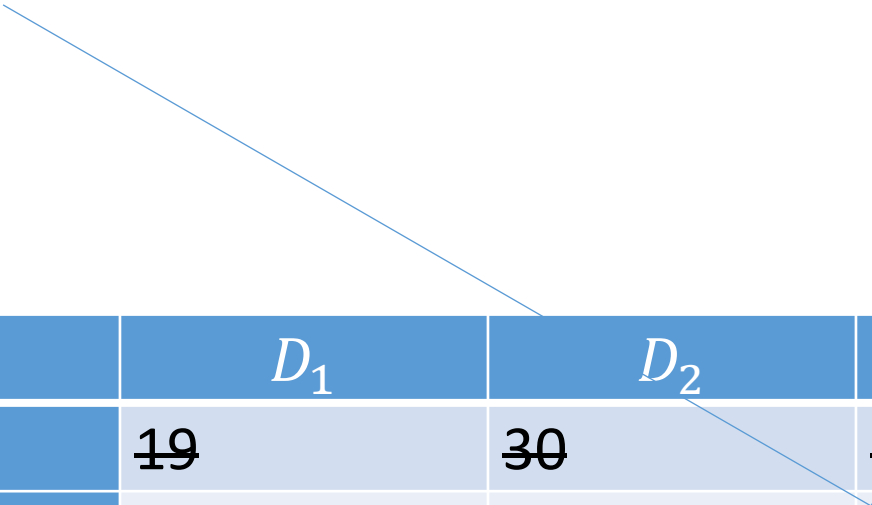
	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10 (7)	7 (0)
S_2	70	30	40	60	9
S_3	40 (3)	8 (8)	70	20 (7)	18 (10) (3) 0
Demand	5 (2)	8 (0)	7	14 (7) 0	

Least Cost



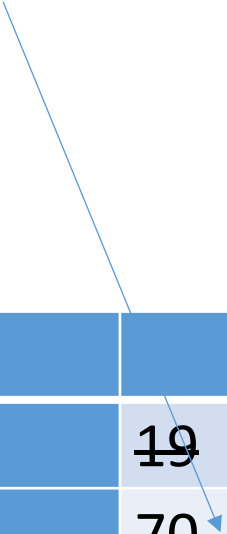
	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10 (7)	7 (0)
S_2	70	30	40	60	9
S_3	40 (3)	8 (8)	70	20 (7)	18 (10) (3) 0
Demand	5 (2)	8 (0)	7	14 (7) 0	

Least Cost



	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10 (7)	7 (0)
S_2	70	30	40 (7)	60	9 (2)
S_3	40 (3)	8 (8)	70	20 (7)	18 (10) (3) 0
Demand	5 (2)	8 (0)	7 (0)	14 (7) 0	

Least Cost



	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10 (7)	7 (0)
S_2	70 (2)	30	40 (7)	60	9 (2)
S_3	40 (3)	8 (8)	70	20 (7)	18 (10) (3) 0
Demand	5 (2)	8 (0)	7 (0)	14 (7) 0	

Least Cost

	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10 (7)	7 (0)
S_2	70 (2)	30	40 (7)	60	9 (2) 0
S_3	40 (3)	8 (8)	70	20 (7)	18 (10) (3) 0
Demand	5 (2) 0	8 (0)	7 (0)	14 (7) 0	

Total Cost= $10 \times 7 + 70 \times 2 + 40 \times 7 + 40 \times 3 + 8 \times 8 + 20 \times 7 = 814$

1.North-West Corner method

	D_1	D_2	D_3	D_4	Supply
S_1	18	8	4	3	500
S_2	12	14	20	2	400
S_3	6	9	23	25	300
Demand	250	350	600	150	

1. North-West Corner method

	D_1	D_2	D_3	D_4	Supply
S_1	18	8	4	3	500
S_2	12	14	20	2	400
S_3	6	9	23	25	300
Demand	250	350	600	150	

$$\text{Total Supply} = \sum_{i=1}^3 a_i = 500 + 400 + 300 = 1200$$

$$\text{Total Demand} = \sum_{j=1}^4 b_j = 250 + 350 + 600 + 150 = 1350$$

Total Supply $\neq$ Total Demand

1.North-West Corner method

	D_1	D_2	D_3	D_4	Supply
S_1	18	8	4	3	500
S_2	12	14	20	2	400
S_3	6	9	23	25	300
Demand	250	350	600	150	

Introduce dummy row

	D_1	D_2	D_3	D_4	Supply
S_1	18	8	4	3	500
S_2	12	14	20	2	400
S_3	6	9	23	25	300
	0	0	0	0	150
Dem and	250	350	600	150	

	D_1	D_2	D_3	D_4	Supply
S_1	18 (250)	8	4	3	500 (250)
S_2	12	14	20	2	400
S_3	6	9	23	25	300
	0	0	0	0	150
Dem and	250-0	350	600	150	

	D_1	D_2	D_3	D_4	Supply
S_1	18 (250)	8(250)	4	3	500 (250) - 0
S_2	12	14	20	2	400
S_3	6	9	23	25	300
	0	0	0	0	150
Dem and	250-0	350(100)	600	150	

	D_1	D_2	D_3	D_4	Supply
S_1	18 (250)	8(250)	4	3	500 (250) 0
S_2	12	14(100)	20	2	400 (300)
S_3	6	9	23	25	300
	0	0	0	0	150
Dem and	250 0	350(100 + 0)	600	150	

	D_1	D_2	D_3	D_4	Supply
S_1	18 (250)	8(250)	4	3	500 (250) 0
S_2	12	14(100)	20(300)	2	400 (300) 0
S_3	6	9	23	25	300
	0	0	0	0	150
Dem and	250 0	350 (100) 0	600 (300)	150	

	D_1	D_2	D_3	D_4	Supply
S_1	18 (250)	8(250)	4	3	500 (250) 0
S_2	12	14(100)	20(300)	2	400 (300) 0
S_3	6	9	23(300)	25	300 0
	0	0	0	0	150
Dem and	250 0	350 (100) 0	600 (300) 0	150	

	D_1	D_2	D_3	D_4	Supply
S_1	18 (250)	8(250)	4	3	500 (250) 0
S_2	12	14(100)	20(300)	2	400 (300) 0
S_3	6	9	23(300)	25	300 0
	0	0	0	0(150)	150 0
Dem and	250 0	350 (100) 0	600 (300) 0	150 0	

Total Cost= 18x250
+8x250
+14x100+20x300
+23x300+0x150
=23300

Least Cost method

	D_1	D_2	D_3	D_4	Supply
S_1	18	8	4	3	500
S_2	12	14	20	2	400
S_3	6	9	23	25	300
	0	0	0	0	150
Dem and	250	350	600	150	

	D_1	D_2	D_3	D_4	Supply
S_1	18	8	4	3	500
S_2	12	14	20	2	400
S_3	6	9	23	25	300
	0	0	0	0	150
Dem and	250	350	600	150	

Least Cost

	D_1	D_2	D_3	D_4	Supply
S_1	18	8	4	3	500
S_2	12	14	20	2 (150)	400 (250)
S_3	6	9	23	25	300
	0	0	0	0	150
Dem and	250	350	600	150 (150)	

Least Cost

	D_1	D_2	D_3	D_4	Supply
S_1	18	8	4(500)	3	500 0
S_2	12	14	20	2 (150)	400(250)
S_3	6	9	23	25	300
	0	0	0	0	150
Dem and	250	350	600 (100)	150 (150)	

Least Cost

	D_1	D_2	D_3	D_4	Supply
S_1	18	8	4(500)	3	500 0
S_2	12	14	20	2 (150)	400 (250)
S_3	6 (250)	9	23	25	300 (50)
	0	0	0	0	150
Dem and	250 (0)	350	600 (100)	150 (150)	

Least Cost

	D_1	D_2	D_3	D_4	Supply
S_1	18	8	4(500)	3	500 0
S_2	12	14	20	2 (150)	400(250)
S_3	6 (250)	9 (50)	23	25	300 (50) 0
	0	0	0	0	150
Dem and	250 (0)	350 (300)	600 (100)	150 (150)	

Least Cost

	D_1	D_2	D_3	D_4	Supply
S_1	18	8	4(500)	3	500 0
S_2	12	14 (250)	20	2 (150)	400(250) 0
S_3	6 (250)	9 (50)	23	25	300 (50) 0
	0	0	0	0	150
Dem and	250 (0)	350 (300) 50	600 (100)	150 (150)-0	

Least Cost

	D_1	D_2	D_3	D_4	Supply
S_1	18	8	4(500)	3	500 0
S_2	12	14 (250)	20	2 (150)	400 (250) 0
S_3	6 (250)	9 (50)	23	25	300 (50) 0
	0	0 (50)	0 (100)	0	150 0
Dem and	250 (0)	350 (300) 50 0	600 (100) 0	150 (150) 0	

Total Cost= $4 \times 500 + 14 \times 250 + 2 \times 150 + 6 \times 250 + 9 \times 50 = 7750$

Optimal Solution

Vogel's Approximation Method (VAM)

Step1: Find the difference (penalty) between the smallest and next smallest costs in each row (column) and write them in brackets against the corresponding row (column).

Step 2: Identify the row (or) column with largest penalty. If a tie occurs break the tie arbitrarily. Choose the cell with smallest cost in that selected row or column and allocate as much as possible to Choose the cell with smallest cost in that selected row or column and allocate as much as possible to this cell and cross out the satisfied row or column and go to step (3).

Step 3: Again compute the column and row penalties for the reduced transportation table and then go to step (2). Repeat the procedure until all the rim requirements are satisfied.

	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	

	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	

Calculate the slack cost (penalty value) for each row and column

difference (penalty) between the smallest and next smallest costs in each row

	D_1	D_2	D_3	D_4	Supply	P_1
S_1	19	30	50	10	7	9
S_2	70	30	40	60	9	10
S_3	40	8	70	20	18	10
Demand	5	8	7	14		

	D_1	D_2	D_3	D_4	Supply	P_1
S_1	19	30	50	10	7	9
S_2	70	30	40	60	9	10
S_3	40	8	70	20	18	12
Demand	5	8	7	14		
P_1	21	22	10	10		

Row Penalty

Column
Penalty

difference (penalty) between the smallest and
next smallest costs in each column

	D_1	D_2	D_3	D_4	Supply	P_1
S_1	19	30	50	10	7	9
S_2	70	30	40	60	9	10
S_3	40	8	70	20	18	12
Demand	5	8	7	14		
P_1	21	22	10	10		

Maximum penalty value

Minimum unit cost

The maximum penalty, 22, occurs in column D_2

The minimum c_{ij} in this row is $c_{32} = 8$

	D_1	D_2	D_3	D_4	Supply	P_1
S_1	19	30	50	10	7	9
S_2	70	30	40	60	9	10
S_3	40	8 (8)	70	20	18 10	12
Demand	5	8 0	7	14		
P_1	21	22	10	10		



Maximum penalty value



Minimum unit cost

	D_1	D_2	D_3	D_4	Supply	P_1	P_2
S_1	19 (5)	30	50	10	7 2	9	9
S_2	70	30	40	60	9	10	20
S_3	40	8 (8)	70	20	18 10	12	20
Demand	5 0	8 0	7	14			
P_1	21	22	10	10			
P_2	21	-	10	10			

Maximum penalty value

Minimum unit cost

	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3
S_1	19 (5)	30	50	10	7 2	9	9	40
S_2	70	30	40	60	9	10	20	20
S_3	40	8 (8)	70	20 (10)	18 10 0	12	20	50
Demand	5 0	8 0	7	14 (4)				
P_1	21	22	10	10				
P_2	21	-	10	10				
P_3	-	-	10	10				

Minimum unit cost

Maximum penalty value

	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3	P_4
S_1	19 (5)	30	50	10	7 2	9	9	40	40
S_2	70	30	40	60	9	10	20	20	20
S_3	40	8 (8)	70	20 (10)	18 10 0	12	20	50	-
Demand	50	80	7	14 (4)					
P_1	21	22	10	10					
P_2	21	-	10	10					
P_3	-	-	10	10					
P_4	-	-	10	50					

	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3	P_4
S_1	19 (5)	30	50	10 (2)	7 2 0	9	9	40	40
S_2	70	30	40	60	9	10	20	20	20
S_3	40	8 (8)	70	20 (10)	18 10 0	12	20	50	-
Demand	5 0	8 0	7	14 (4) (2)					
P_1	21	22	10	10					
P_2	21	-	10	10					
P_3	-	-	10	10					
P_4	-	-	10	50					

Minimum unit cost

Maximum penalty value

	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3	P_4
S_1	19 (5)	30	50	10 (2)	7 2 0	9	9	40	40
S_2	70	30	40*7	60* 2	9*2*0	10	20	20	20
S_3	40	8 (8)	70	20 (10)	18 10 0	12	20	50	-
Demand	5-0	8 0	7*0	14 (4) (2)* 0					
P_1	21	22	10	10					
P_2	21	-	10	10					
P_3	-	-	10	10					
P_4	-	-	10	50					

Total Cost=

19x5+10x2+40x7+60x2+8x8+20x10=779 Units

Example 2

Unbalance problem

	D_1	D_2	D_3	D_4	Supply
S_1	18	8	4	3	500
S_2	12	14	20	2	400
S_3	6	9	23	25	300
	0	0	0	0	150
Dem and	250	350	600	150	

	D_1	D_2	D_3	D_4	Supply	P_1
S_1	18	8	4	3	500	1
S_2	12	14	20	2	400	10
S_3	6	9	23	25	300	3
	0	0	0	0	150	0
Demand	250	350	600	150		
P_1	6	8	4	1		

Minimum unit cost

Maximum penalty value

	D_1	D_2	D_3	D_4	Supply	P_1
S_1	18	8	4	3	500	1
S_2	12	14	20	2 * 150	400 * 250	10
S_3	6	9	23	25	300	3
	0	0	0	0	150	0
Demand	250	350	600	150 * 0		
P_1	6	8	4	1		

Minimum unit cost

Maximum penalty value

	D_1	D_2	D_3	D_4	Supply	P_1	P_2
S_1	18	8	4	3	500	1	
S_2	12	14	20	2* 150	400 * 250	10	
S_3	6	9	23	25	300	3	
	0	0	0	0	150	0	
Demand	250	350	600	150 * 0			
P_1	6	8	4	1			
P_2							

	D_1	D_2	D_3	D_4	Supply	P_1	P_2
S_1	18	8	4	3	500	1	4
S_2	12	14	20	2*150	400* 250	10	2
S_3	6	9	23	25	300	3	3
	0	0	0	0	150	0	0
Demand	250	350	600	150 * 0			
P_1	6	8	4	1			
P_2	6	8	4	-			

	D_1	D_2	D_3	D_4	Supply	P_1	P_2
S_1	18	8	4	3	500	1	4
S_2	12	14	20	2*150	400*250	10	2
S_3	6	9	23	25	300	3	3
	0	0*150	0	0	150*0		
Demand	250	350*200	600	150*0			
P_1	6	8	4	1			
P_2	6	8	4	-			

Minimum unit cost

Maximum penalty value

	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3
S_1	18	8	4	3	500	1	4	4
S_2	12	14	20	2*150	400*250	10	2	2
S_3	6	9	23	25	300	3	3	3
	0	0*150	0	0	150*0	0	0	-
Demand	250	350 *200	600	150*0				
P_1	6	8	4	1				
P_2	6	8	4	-				
P_3	6	1	16	-				

	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3
S_1	18	8	4	3	500	1	4	4
S_2	12	14	20	2*150	400*250	10	2	2
S_3	6	9	23	25	300	3	3	3
	0	0*150	0	0	150*0	0	0	-
Demand	250	350 *200	600	150*0				
P_1	6	8	4	1				
P_2	6	8	4	-				
P_3	6	1	16	-				

Minimum unit cost

Maximum penalty value

	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3
S_1	18	8	4* 500	3	500 * 0	1	4	4
S_2	12	14	20	2* 150	400 * 250	10	2	2
S_3	6	9	23	25	300	3	3	3
	0	0 * 150	0	0	150 * 0	0	0	-
Demand	250	350 * 200	600 * 100	150 * 0				
P_1	6	8	4	1				
P_2	6	8	4	-				
P_3	6	1	16	-				

Minimum unit cost

Maximum penalty value

	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3	P_4
S_1	18	8	4* 500	3	500 * 0	1	4	4	-
S_2	12	14	20	2* 150	400 * 250	10	2	2	2
S_3	6	9	23	25	300	3	3	3	3
	0	0 *	0	0	150 * 0	0	0	-	-
		150							
Demand	250	350	600 *	150 * 0					
		* 200	100						
P_1	6	8	4	1					
P_2	6	8	4	-					
P_3	6	1	16	-					
P_4	6	5	3	-					

Minimum unit cost

Maximum penalty value

	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3	P_4
S_1	18	8	4* 500	3	500 * 0	1	4	4	-
S_2	12	14	20	2*150	400* 250	10	2	2	2
S_3	6 *250	9	23	25	300*50	3	3	3	3
	0	0 * 150	0	0	150* 0	0	0	-	-
Demand	250* 0	350 *200	600 * 100	150 * 0					
P_1	6	8	4	1					
P_2	6	8	4	-					
P_3	6	1	16	-					
P_4	6	5	3	-					

Minimum unit cost

Maximum penalty value

[illegible]

	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3	P_4	P_5
S_1	18	8	4* 500	3	500 * 0	1	4	4	-	-
S_2	12	14	20	2* 150	400* 250	10	2	2	2	6
S_3	6 * 250	9	23	25	300* 50	3	3	3	3	14
	0	0 * 150	0	0	150* 0	0	0	-	-	
Demand	250* 0	350 * 200	600 * 100	150 * 0						
P_1	6	8	4	1						
P_2	6	8	4	-						
P_3	6	1	16	-						
P_4	6	5	3	-						
P_5	-	5	3	-						

	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3	P_4	P_5
S_1	18	8	4*500	3	500*0	1	4	4	-	-
S_2	12	14	20	2*150	400*250	10	2	2	2	6
S_3	6	9	23	25	300*50	3	3	3	3	14
	0	0*	0	0	150*0	0	0	-	-	
Demand	250*	350	600*	150*0						
P_1	6	8	4	1						
P_2	6	8	4	-						
P_3	6	1	16	-						
P_4	6	5	3	-						
P_5	-	5	3	-						

Minimum unit cost

Maximum penalty value

	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3	P_4	P_5
S_1	18	8	4* 500	3	500 * 0	1	4	4	-	-
S_2	12	14	20	2* 150	400 * 250	10	2	2	2	6
S_3	6	9 *	23	25	300 * 50 *	3	3	3	3	14
	*250	50			0					
	0	0 *	0	0	150 * 0	0	0	-	-	
		150								
Demand	250*	350	600 *	150 * 0						
	0	*200	100							
		*150								
P_1	6	8	4	1						
P_2	6	8	4	-						
P_3	6	1	16	-						
P_4	6	5	3	-						
P_5	-	5	3	-						

Minimum unit cost

Maximum penalty value

	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3	P_4	P_5	P_6
S_1	18	8	4*500	3	500*0	1	4	4	-	-	-
S_2	12	14*150	20*100	2*150	400*250*100*0	1	2	2	2	6	6
S_3	6	9*50	23	25	300*50*0	3	3	3	3	14	-
	0	0*150	0	0	150*0	0	0	-	-	-	-
Demand	250*0	350*200*150	600*100	150*0							
P_1	6	8	4	1							
P_2	6	8	4	-							
P_3	6	1	16	-							
P_4	6	5	3	-							
P_5	-	5	3	-							
P_6	-										

Total Cost=4x500+150x14+20x100+2x150+6x250+9x50+150x0=6550

Modified Distribution Method

The Modified Distribution Method, also known as MODI method or u-v method, which provides a minimum cost solution (optimal solution) to the transportation problem. The following are the steps involved in this method.

Step 1: Find out the basic feasible solution of the transportation problem using any one of the three methods discussed in the previous section.

Step 2: Introduce dual variables corresponding to the row constraints and the column constraints. If there are m origins and n destinations then there will be $m + n$ dual variables. The dual variables corresponding to the row constraints are represented by $u_i, i = 1, 2, \dots, m$ whereas the dual variables corresponding to the column constraints are represented by $v_j, j = 1, 2, \dots, n$. The values of the dual variables are calculated from the equation given below $u_i + v_j = c_{ij}$ if $x_{ij} > 0$

Step 3: Any basic feasible solution has $m + n - 1$. Thus, there will be $m + n - 1$ equation to determine $m + n$ dual variables. One of the dual variables can be chosen arbitrarily. It is also to be noted that as the primal constraints are equations, the dual variables are unrestricted in sign.

Step 4: If $x_{ij} = 0$, the dual variables calculated in Step 3 are compared with the c_{ij} values of this allocation as $c_{ij} - u_i - v_j$. If all $c_{ij} - u_i - v_j \geq 0$, then by the theorem of complementary slackness it can be shown that the corresponding solution of the transportation problem is optimum. If one or more $c_{ij} - u_i - v_j < 0$, we select the cell with the least value of $c_{ij} - u_i - v_j$ and allocate as much as possible subject to the row and column constraints. The allocations of the number of adjacent cell are adjusted so that a basic variable becomes non-basic.

Step 5: A fresh set of dual variables are calculated and repeat the entire procedure from Step 1 to Step 5.

Example

Consider the transportation problem given below:

						Supply
	1	9	13	36	51	50
	24	12	16	20	1	100
	14	33	1	23	26	150
Demand	100	70	50	40	40	

Step 1: First we have to determine the basic feasible solution. The basic feasible solution using least cost method is

						Supply
	1*50	9	13	36	51	50
	24	12*60	16	20	1*40	100*60
	14*50	33*10	1*50	23*40	26	150*100
Demand	100*50	70	50	40	40	

$x_{11} = 50, x_{22} = 60, x_{25} = 40, x_{31} = 50, x_{32} = 10, x_{33} = 50 \text{ and } x_{34} = 40$

						Supply
	1*50	9	13	36	51	50
	24	12*60	16	20	1*40	100
	14*50	33*10	1*50	23*40	26	150
Demand	100	70	50	40	40	

Step 2: The dual variables u_1, u_2, u_3 and v_1, v_2, v_3, v_4, v_5 can be calculated from the corresponding c_{ij} values, that is

$$u_1 + v_1 = 1, u_2 + v_2 = 12, u_2 + v_5 = 1, u_3 + v_1 = 14, u_3 + v_2 = 33, u_3 + v_3 = 1, u_3 + v_4 = 23$$

						Supply
	1*50	9	13	36	51	50
	24	12*60	16	20	1*40	100
	14*50	33*10	1*50	23*40	26	150
Demand	100	70	50	40	40	

Step 3: Choose one of the dual variables arbitrarily is zero that is $u_3 = 0$ as it occurs most often in the above equations. The values of the variables calculated are

$$u_1 + v_1 = 1, u_2 + v_2 = 12, u_2 + v_5 = 1, v_1 = 14, v_2 = 33, v_3 = 1, v_4 = 23$$

$$u_1 = -13, u_2 = -21, v_5 = 22, v_1 = 14, v_2 = 33, v_3 = 1, v_4 = 23$$

						Supply
	1*50	9	13	36	51	50
	24	12*60	16	20	1*40	100
	14*50	33*10	1*50	23*40	26	150
Demand	100	70	50	40	40	

$$u_1 = -13, u_2 = -21, u_3 = 0, v_5 = 22, v_1 = 14, v_2 = 33, v_3 = 1, v_4 = 23$$

Step 4: Now we calculate $c_{ij} - u_i - v_j$ values for all the cells where $x_{ij} = 0$ (i.e. unallocated cell by the basic feasible solution)

That is

$$Cell(1,2) = c_{12} - u_1 - v_2 = 9 + 13 - 33 = -11$$

$$Cell(1,3) = c_{13} - u_1 - v_3 = 13 + 13 - 1 = 25$$

$$Cell(1,4) = c_{14} - u_1 - v_4 = 36 + 13 - 23 = 26$$

$$Cell(1,5) = c_{15} - u_1 - v_5 = 51 + 13 - 22 = 42$$

$$Cell(2,1) = c_{21} - u_2 - v_1 = 24 + 21 - 14 = 31$$

$$Cell(2,3) = c_{23} - u_2 - v_3 = 16 + 21 - 1 = 36$$

$$Cell(2,4) = c_{24} - u_2 - v_4 = 20 + 21 - 23 = 18$$

$$Cell(3,5) = c_{35} - u_3 - v_5 = 26 - 0 - 22 = 4$$

Note that in the above calculation all the $c_{ij} - u_i - v_j \geq 0$ except for cell (1, 2) where $c_{12} - u_1 - v_2 = 9 + 13 - 33 = -11$.

Thus in the next iteration x_{12} will be a basic variable changing one of the present basic variables non-basic. We also observe that for allocating one unit in cell (1, 2) we have to reduce one unit in cells (3, 2) and (1, 1) and increase one unit in cell (3, 1). The net transportation cost for each

unit of such reallocation is $-33 - 1 + 9 + 14 = -11$

The maximum that can be allocated to cell (1, 2) is 10 otherwise the allocation in the cell (3, 2) will be negative. Thus, the revised basic feasible solution is

$$x_{11} = 50, x_{22} = 60, x_{25} = 40, x_{31} = 50, x_{32} = 10, x_{33} = 50 \text{ and } x_{34} = 40$$

$$x_{11}=40, x_{12}=10, x_{22}=60, x_{25}=40, x_{31}=60, x_{33}=50, x_{34}=40$$

1*50(-)	9 (+)
24	12*60
14*50(+)	33*10 (-)

						Supply
	1*50	9	13	36	51	50
	24	12*60	16	20	1*40	100
	14*50	33*10	1*50	23*40	26	150
Demand	100	70	50	40	40	

$$x_{11} = 50, x_{22} = 60, x_{25} = 40, \\ x_{31} = 50, x_{32} = 10, x_{33} = 50 \\ \text{and } x_{34} = 40$$

1*50(-)	9 (+)
24	12*60
14*50(+)	33*10 (-)

1*50(-)- 10	9 (+)
24	12*60
14*50(+)	33*10 (-)

New cost

$$x_{11} = 40, x_{12} = 10, x_{22} = 60, x_{25} = 40, \\ x_{31} = 60, x_{32} = 10, x_{33} = 50 \\ \text{and } x_{34} = 40$$

						Supply
	1*40	9(+10=10)	13	36	51	50
	24	12*60	16	20	1*40	100
	14*60	33	1*50	23*40	26	150
Demand	100	70	50	40	40	

				Supply
	8	9	7	40
	4	3	5	25
	8	5	6	35
Demand	30	30	40	

NWCM

				Supply
	8*30	9*10	7	40
	4	3*20	5*5	25
	8	5	6*35	35
Demand	30	30	40	

$$x_{11} = 30, x_{12} = 10, x_{22} = 20, \\ x_{23} = 5, x_{33} = 35$$

$$\text{Total Cost} = 8 \times 30 + 9 \times 10 + 3 \times 20 + 5 \times 5 + 6 \times 35 = 625$$

NWCM

				Supply
	8*30	9*10	7	40
	4	3*20	5*5	25
	8	5	6*35	35
Demand	30	30	40	

$$x_{11} = 30, x_{12} = 10, x_{22} = 20, \\ x_{23} = 5, x_{33} = 35$$

The dual variables u_1, u_2, u_3 and v_1, v_2, v_3 , can be calculated from the corresponding c_{ij} values, that is

$$u_1 + v_1 = 8, u_1 + v_2 = 9, u_2 + v_2 = 3, u_2 + v_3 = 5, u_3 + v_3 = 6$$

Choose any one of the dual variables arbitrarily zero. Let $u_1=0$

$$v_1 = 8, v_2 = 9, u_2 = -9, v_3 = 14, u_3 = -8$$

Lest Cost method

				Supply
	8*30	9	7*10	40
	4	3*25	5	25
	8	5*5	6*30	35
Demand	30	30	40	

$$x_{11} = 30, x_{13} = 10, x_{22} = 25, \\ x_{32} = 5, x_{33} = 30$$

Modimethod

				Supply
	8*30	9	7*10	40
	4	3*25	5	25
	8	5*5	6*30	35
Demand	30	30	40	

$$x_{11} = 40, x_{13} = 10, x_{22} = 25, \\ x_{32} = 5, x_{33} = 30$$

The dual variables u_1, u_2, u_3 and v_1, v_2, v_3 , can be calculated from the corresponding c_{ij} values, that is

$$u_1 + v_1 = 8, u_1 + v_3 = 7, u_2 + v_2 = 3, u_3 + v_2 = 5, u_3 + v_3 = 6$$

Choose any one of the dual variables arbitrarily zero. Let $u_1 = 0$

$$v_1 = 8, v_2 = 6, u_2 = -3, v_3 = 7, u_3 = -1$$

Modi method

	$v_1=8$	$v_2=6$	$v_3=7$	Supply
$u_1=0$	8*30	9	7*10	40
$u_2=-3$	4	3*25	5	25
$u_3=-1$	8	5*5	6*30	35
Demand	30	30	40	

$$x_{11} = 40, x_{13} = 10, x_{22} = 25, \\ x_{32} = 5, x_{33} = 30$$

The dual variables u_1, u_2, u_3 and v_1, v_2, v_3 , can be calculated from the corresponding c_{ij} values, that is

$$u_1 + v_1 = 8, u_1 + v_3 = 7, u_2 + v_2 = 3, u_3 + v_2 = 5, u_3 + v_3 = 6$$

Choose any one of the dual variables arbitrarily zero. Let $u_1=0$

$$v_1 = 8, v_2 = 6, u_2 = -3, v_3 = 7, u_3 = -1$$

Modi method

	$v_1=8$	$v_2=6$	$v_3=7$	Supply
$u_1=0$	8*30	9	7*10	40
$u_2=-3$	4	3*25	5	25
$u_3=-1$	8	5*5	6*30	35
Demand	30	30	40	

Now we calculate $c_{ij} - u_i - v_j$ values for all the cells where $x_{ij} = 0$ (.e. unallocated cell by the basic feasible solution)
That is

$$\begin{aligned}
 \text{Cell}(1,2) &= c_{12} - u_1 - v_2 = 9 - 0 - 6 = 3 \\
 \text{Cell}(2,1) &= c_{21} - u_2 - v_1 = 4 + 3 - 8 = -1 \\
 \text{Cell}(2,3) &= c_{23} - u_2 - v_3 = 5 + 3 - 7 = 1 \\
 \text{Cell}(3,1) &= c_{31} - u_3 - v_1 = 8 + 1 - 8 = 1
 \end{aligned}$$

Note that in the above calculation all the $c_{ij} - u_i - v_j \geq 0$ except for cell (2,1) where $c_{21} - u_2 - v_1 = 4 + 3 - 8 = -1$.

Modi method

	$v_1=8$	$v_2=6$	$v_3=7$	Supply
$u_1=0$	8*30	9	7*10	40
$u_2= -3$	4	3*25	5	25
$u_3=-1$	8	5*5	6*30	35
Demand	30	30	40	

Note that in the above calculation all the $c_{ij} - u_i - v_j \geq 0$ except for cell (2,1) where $c_{21} - u_2 - v_1 = 4 + 3 - 8 = -1$.

Modi method

	$v_1=8$	$v_2=6$	$v_3=7$	Supply
$u_1=0$	8*30-	9	7*10+	40
$u_2=-3$	4+	3*25-	5	25
$u_3=-1$	8	5*5+	6*30-	35
Demand	30	30	40	

Note that in the above calculation all the $c_{ij} - u_i - v_j \geq 0$ except for cell (2,1) where $c_{21} - u_2 - v_1 = 4 + 3 - 8 = -1$.

Thus in the next iteration x_{21} will be a basic variable changing one of the present basic variables non-basic.

Select the minimum value from allocated values to the cell with a minus sign. Subtract this value from the cell with a minus sign and add to the cell with a plus sign.

Modi method

	$v_1=8$	$v_2=6$	$v_3=7$	Supply
$u_1=0$	8*30-+(-25)	9	7*10 ++(+25)	40
$u_2=-3$	4 ++(+25)	3*25-+(-25)	5	25
$u_3=-1$	8	5*5++(+25)	6*30 -+(-25)	35
Demand	30	30	40	

Select the minimum value from allocated values to the cell with a minus sign. Subtract this value from the cell with a minus sign and add to the cell with a plus sign.

MODImethod

	$v_1=8$	$v_2=6$	$v_3=7$	Supply
$u_1=0$	8*5	9	7*35	40
$u_2=-3$	4 25	3	5	25
$u_3=-1$	8	5*30	6*5	35
Demand	30	30	40	

Now need to check whether this solution is best

The dual variables u_1, u_2, u_3 and v_1, v_2, v_3 , can be calculated from the corresponding c_{ij} values, that is

$$u_1 + v_1 = 8, u_1 + v_3 = 7, u_2 + v_1 = 4, u_3 + v_2 = 5, u_3 + v_3 = 6$$

Choose any one of the dual variables arbitrarily zero. Let $u_1=0$

$$v_1 = 8, v_2 = 6, u_2 = -4, v_3 = 7, u_3 = -1$$

MODImethod

	$v_1=8$	$v_2=6$	$v_3=7$	Supply
$u_1=0$	8*5	9	7*35	40
$u_2=-3$	4 25	3	5	25
$u_3=-1$	8	5*30	6*5	35
Demand	30	30	40	

Now need to check whether this solution is best

Now we calculate $c_{ij} - u_i - v_j$ values for all the cells where $x_{ij} = 0$ (.e. unallocated cell by the basic feasible solution)

That is

$$\text{Cell}(1,2) = c_{12} - u_1 - v_2 = 9 - 0 - 6 = 3$$

$$\text{Cell}(2,2) = c_{22} - u_2 - v_2 = 3 + 4 - 6 = 1$$

$$\text{Cell}(2,3) = c_{23} - u_2 - v_3 = 4 + 3 - 7 = 0$$

$$\text{Cell}(3,1) = c_{31} - u_3 - v_1 = 8 + 1 - 8 = 1$$

$$v_1 = 8, v_2 = 6, u_2 = -4, v_3 = 7, u_3 = -1$$

MODImethod

	$v_1=8$	$v_2=6$	$v_3=7$	Supply
$u_1=0$	8*5	9	7*35	40
$u_2=-3$	4 25	3	5	25
$u_3=-1$	8	5*30	6*5	35
Demand	30	30	40	

Now need to check whether this solution is best

Now we calculate $c_{ij} - u_i - v_j$ values for all the cells where $x_{ij} = 0$ (.e. unallocated cell by the basic feasible solution)

That is

$$\text{Cell}(1,2) = c_{12} - u_1 - v_2 = 9 - 0 - 6 = 3$$

$$\text{Cell}(2,2) = c_{22} - u_2 - v_2 = 3 + 4 - 6 = 1$$

$$\text{Cell}(2,3) = c_{23} - u_2 - v_3 = 4 + 3 - 7 = 0$$

$$\text{Cell}(3,1) = c_{31} - u_3 - v_1 = 8 + 1 - 8 = 1$$

$$c_{ij} - u_i - v_j \text{ values} \geq 0$$

Hence solution is optimal.

	$v_1=8$	$v_2=6$	$v_3=7$	Supply
$u_1=0$	8*5	9	7*35	40
$u_2=-3$	4 25	3	5	25
$u_3=-1$	8	5*30	6*5	35
Demand	30	30	40	

Optimal Cost= $8 \times 5 + 7 \times 35 + 4 \times 25 + 5 \times 30 + 6 \times 5 = 565$